

Vitamin and mineral supplements in the primary prevention of cardiovascular disease and cancer: An updated systematic evidence review for the U.S.

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BACKGROUND: Vitamin and mineral supplements are commonly used to prevent chronic diseases. **PURPOSE:** To systematically review evidence for the benefit and harms of vitamin and mineral supplements in community-dwelling, nutrient-sufficient adults for the primary prevention of cardiovascular disease (CVD) and cancer. **DATA SOURCES:** MEDLINE, Embase, Cochrane Central Register of Controlled Trials, Cochrane Database of Systematic Reviews, and Database of Systematic Reviews of Effects were searched from January 2005 to 29 January 2013, with manual searches of reference lists and gray literature. **STUDY SELECTION:** Two investigators independently selected and reviewed fair- and good-quality trials for benefit and fair- and good-quality trials and observational studies for harms. **DATA EXTRACTION:** Dual quality assessments and data abstraction. **DATA SYNTHESIS:** Two large trials ($n = 27\,658$) reported lower cancer incidence in men taking a multivitamin for more than 10 years (pooled unadjusted relative risk, 0.93 [95% CI, 0.87 to 0.99]). The study that included women showed no effect in that group. High-quality studies ($k = 24$; $n = 324\,653$) of single and paired nutrients (such as vitamins A, C, or D; folic acid; selenium; or calcium) were scant and heterogeneous and showed no clear evidence of benefit or harm. Neither vitamin E nor β -carotene prevented CVD or cancer, and β -carotene increased lung cancer risk in smokers. **LIMITATIONS:** The analysis included only primary prevention studies in adults without known nutritional deficiencies. Studies were conducted in older individuals and included various supplements and doses under the set upper tolerable limits. Duration of most studies was less than 10 years. **CONCLUSION:** Limited evidence supports any benefit from vitamin and mineral supplementation for the prevention of cancer or CVD. Two trials found a small, borderline-significant benefit from multivitamin supplements on cancer in men only and no effect on CVD. **PRIMARY FUNDING SOURCE:** Agency for Healthcare Research and Quality.